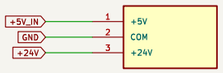
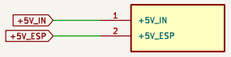
DRAFT — LOW-VOLTAGE DC ONLY — NOT FOR FABRICATION

All connector pin orders are PROJECT-DEFINED. Cable by signal names, not by assuming module pin order.

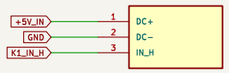
J1
DC input terminal



JP1
removable external-5V supply link



J6
K1 external relay CONTROL only



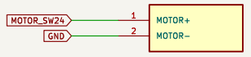
J7
K2 external relay CONTROL only



J8
K2 external relay DC contact cable terminal



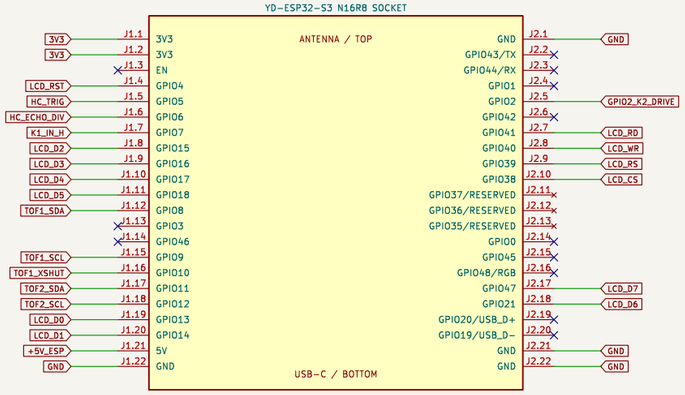
J9
24V motor cable terminal



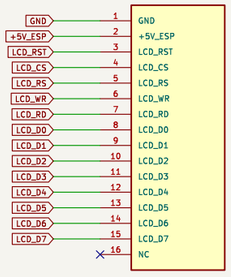
D1
1N540B



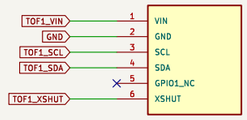
A1
YD-ESP32-S3 N16R8 SOCKET



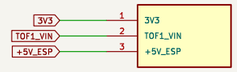
J2
LCD cable header 1x16



J3
ToF1 six-wire cable header



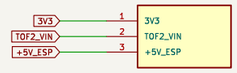
JP2
ToF1 VIN selection jumper



J4
ToF2 four-wire cable header



JP3
ToF2 VIN selection jumper

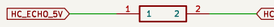


A1: 2x22 YD socket; V1.5 pad audit and mechanical fit pending. GPIO1 and PSRAM-reserved GPIO35/36/37 NC.

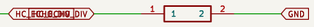
J5
HC-SR04 four-wire cable header



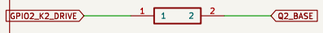
R4
10k



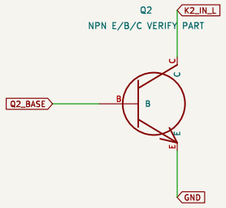
R5
20k



R1
1.5k



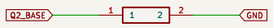
Q2
NPN E/B/C VERIFY PART



R3
4.7k



R2
47k



H1
M2.5 clearance — verify case

H2
M2.5 clearance — verify case

H3
M2.5 clearance — verify case

JP1: remove external 5V link for USB power. JP2/JP3: NO SHUNT until ToF VIN/pull-up voltage verified.

K1 and K2 modules remain external. J8 is 24V DC relay COM/NO only. NO 110V, N or PE on this PCB.

Q2: verify E/B/C before assembly. D1 band K to switched motor+. Existing D-50B 24V/1A startup capacity unverified.

Sheet: /
File: esp32s3_24v_carrier.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/1